

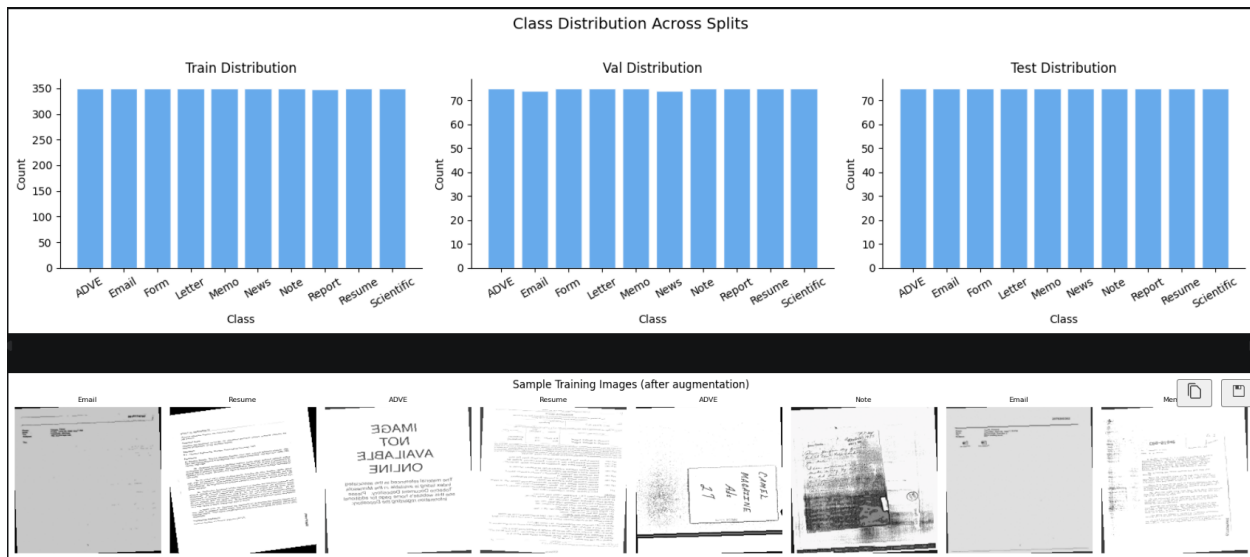


Figure 1: Class Distribution Across Train, Val, and Test Splits with Sample Training Images after Augmentation

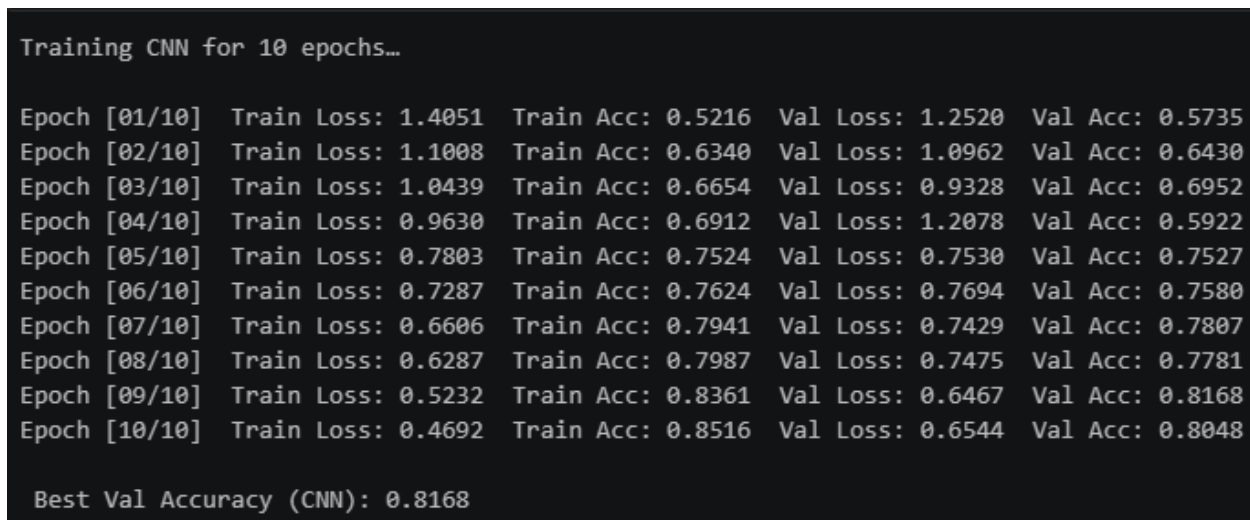


Figure 2: CNN Training Log: Epoch-by-Epoch Train and Validation Loss/Accuracy — Best Val Accuracy: 0.8168

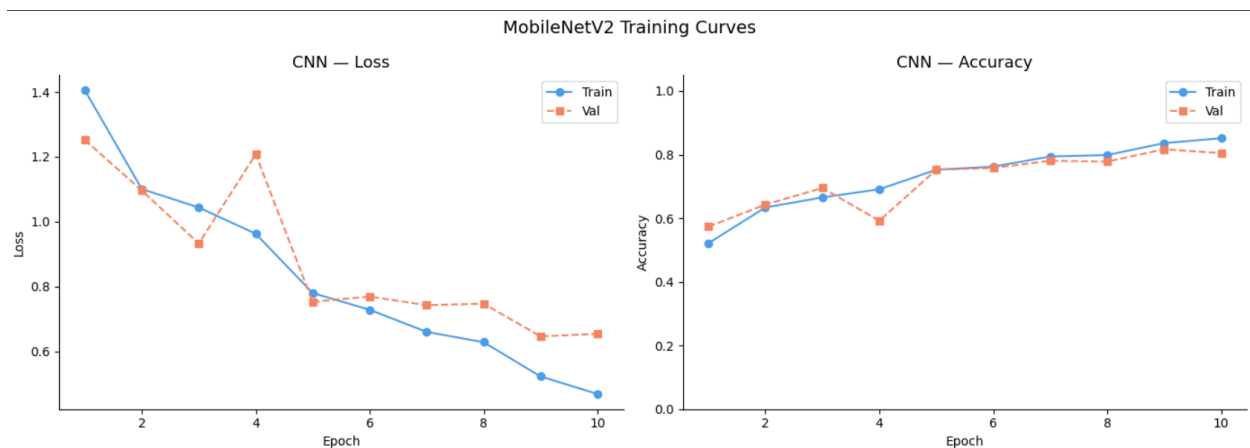


Figure 3: MobileNetV2 (CNN) Training Curves: Loss and Accuracy over 10 Epochs (Train vs Val)

Training MLP for 10 epochs...

Epoch [01/10]	Train Loss: 1.1018	Train Acc: 0.6484	Val Loss: 1.7496	Val Acc: 0.3835
Epoch [02/10]	Train Loss: 0.2357	Train Acc: 0.9677	Val Loss: 0.7660	Val Acc: 0.7507
Epoch [03/10]	Train Loss: 0.0479	Train Acc: 0.9938	Val Loss: 0.7615	Val Acc: 0.7552
Epoch [04/10]	Train Loss: 0.0196	Train Acc: 0.9988	Val Loss: 0.7881	Val Acc: 0.7522
Epoch [05/10]	Train Loss: 0.0113	Train Acc: 0.9994	Val Loss: 0.7925	Val Acc: 0.7522
Epoch [06/10]	Train Loss: 0.0070	Train Acc: 0.9997	Val Loss: 0.8023	Val Acc: 0.7566
Epoch [07/10]	Train Loss: 0.0083	Train Acc: 0.9991	Val Loss: 0.8199	Val Acc: 0.7478
Epoch [08/10]	Train Loss: 0.0100	Train Acc: 0.9984	Val Loss: 0.8243	Val Acc: 0.7507
Epoch [09/10]	Train Loss: 0.0066	Train Acc: 0.9991	Val Loss: 0.8476	Val Acc: 0.7404
Epoch [10/10]	Train Loss: 0.0051	Train Acc: 0.9991	Val Loss: 0.8422	Val Acc: 0.7389

Figure 4: OCR-MLP Training Log: Epoch-by-Epoch Train and Validation Loss/Accuracy over 10 Epochs

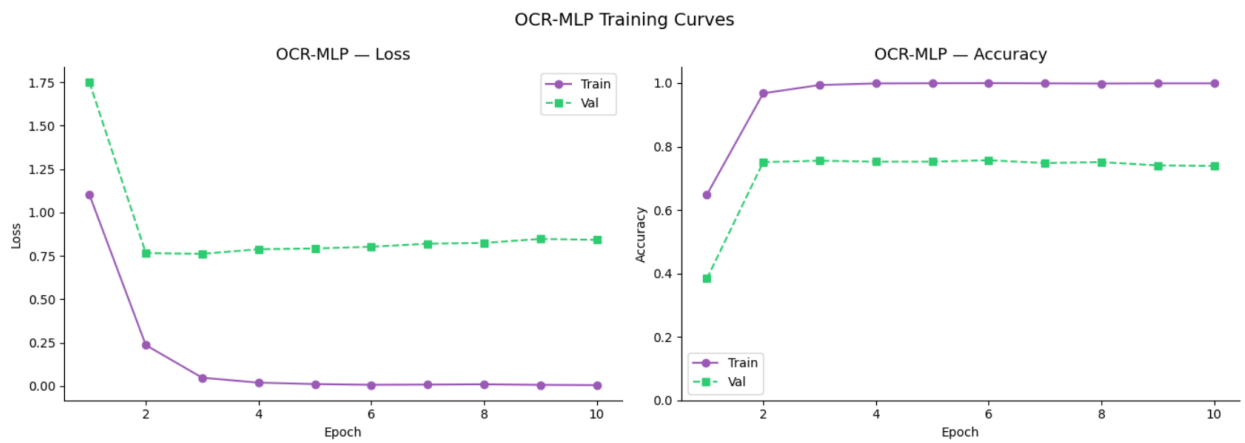


Figure 5: OCR-MLP Training Curves: Loss and Accuracy over 10 Epochs (Train vs Val) — showing Overfitting

CNN (MobileNetV2) – Classification Report				
	precision	recall	f1-score	support
ADVE	0.86	0.87	0.86	75
Email	0.91	0.92	0.91	75
Form	0.76	0.76	0.76	75
Letter	0.84	0.75	0.79	75
Memo	0.78	0.81	0.80	75
News	0.87	0.73	0.80	75
Note	0.87	0.89	0.88	75
Report	0.61	0.71	0.65	75
Resume	0.80	0.89	0.84	75
Scientific	0.87	0.77	0.82	75
accuracy			0.81	750
macro avg	0.82	0.81	0.81	750
weighted avg	0.82	0.81	0.81	750
OCR-MLP – Classification Report				
	precision	recall	f1-score	support
...				
macro avg	0.74	0.73	0.73	687
weighted avg	0.74	0.74	0.74	687

Figure 6: CNN (MobileNetV2) and OCR-MLP Classification Reports: Per-Class Precision, Recall, F1-Score

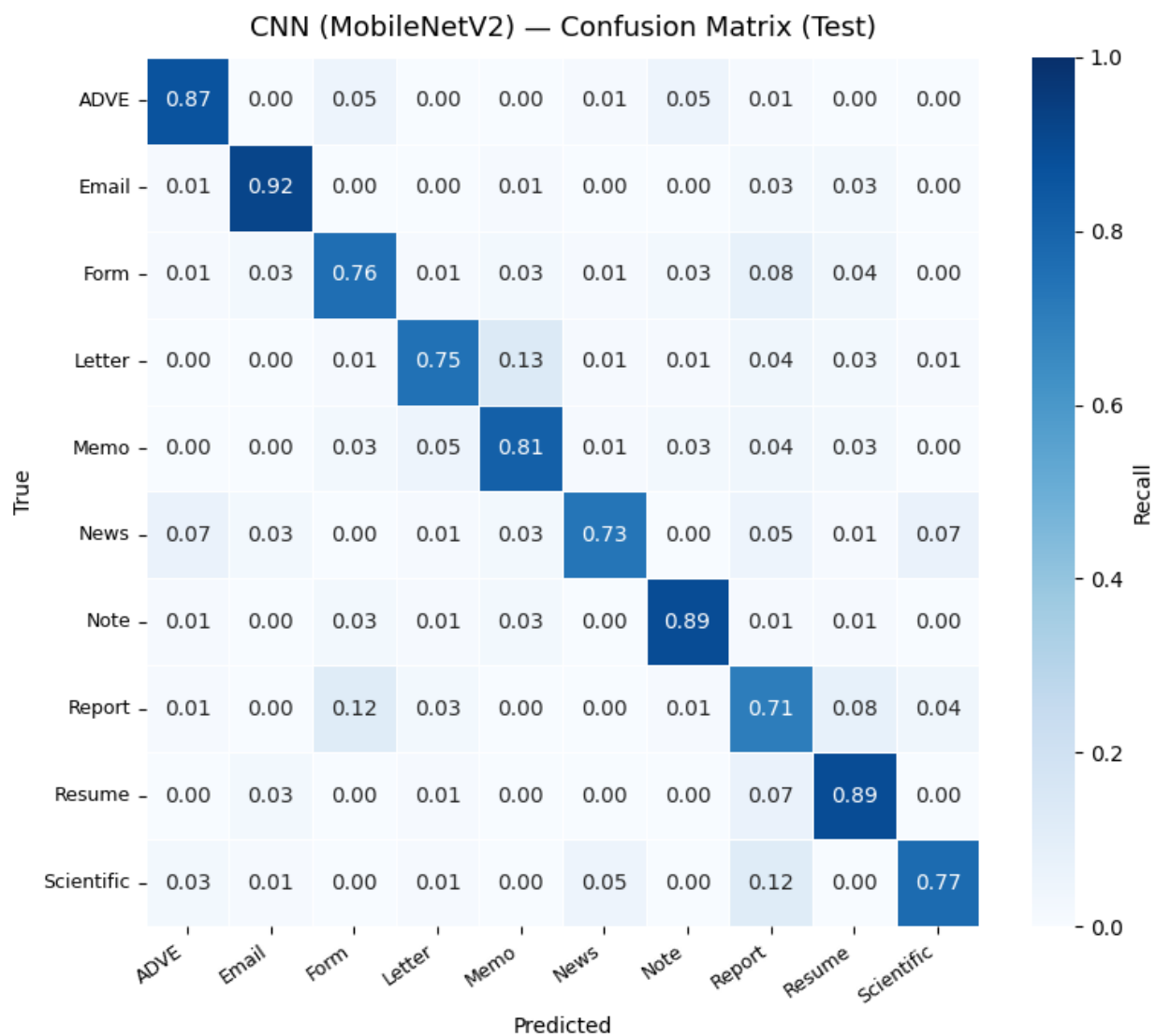


Figure 7: CNN (MobileNetV2) Confusion Matrix on Test Set (10 Document Classes)

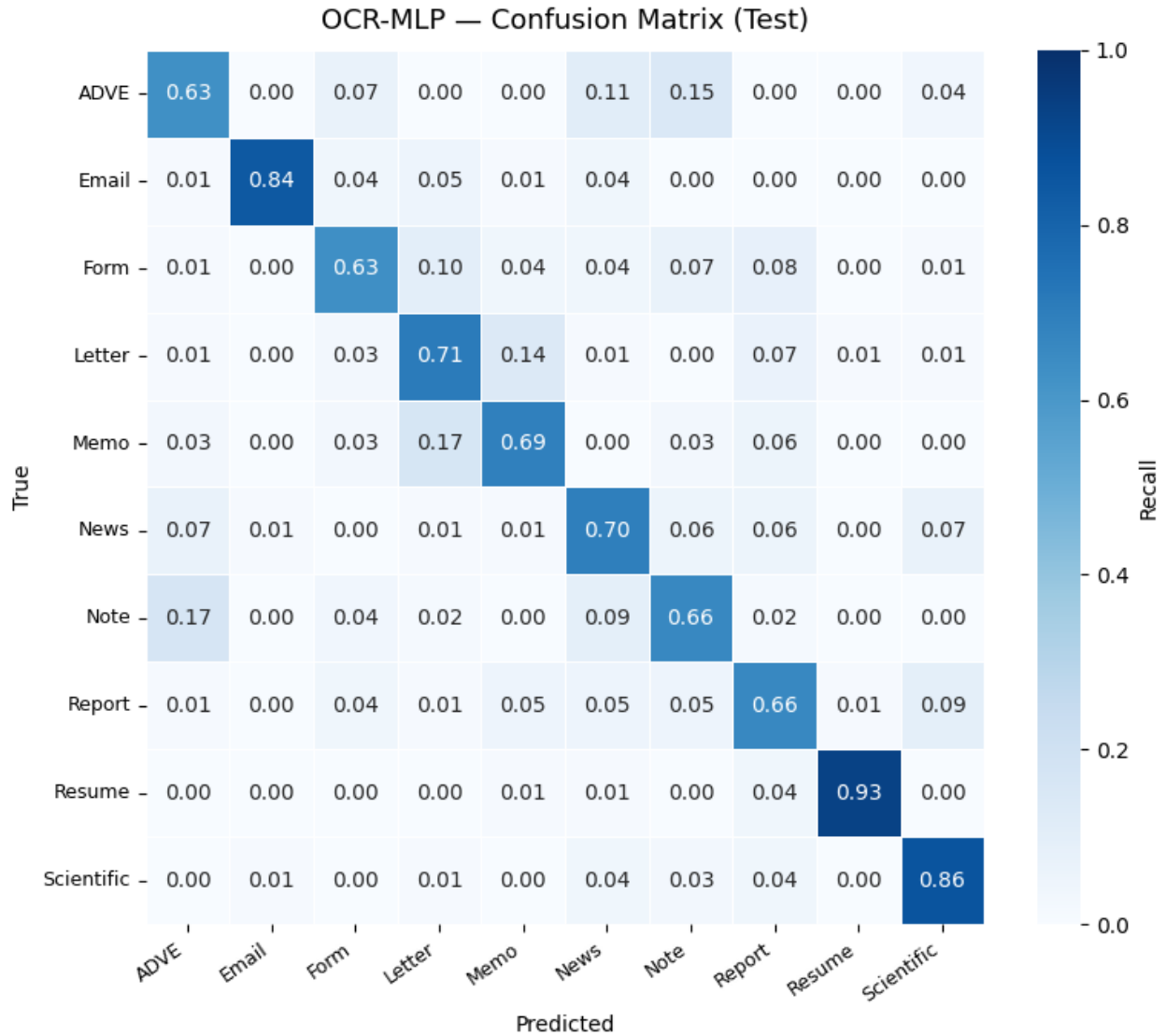


Figure 8: OCR-MLP Confusion Matrix on Test Set (10 Document Classes)

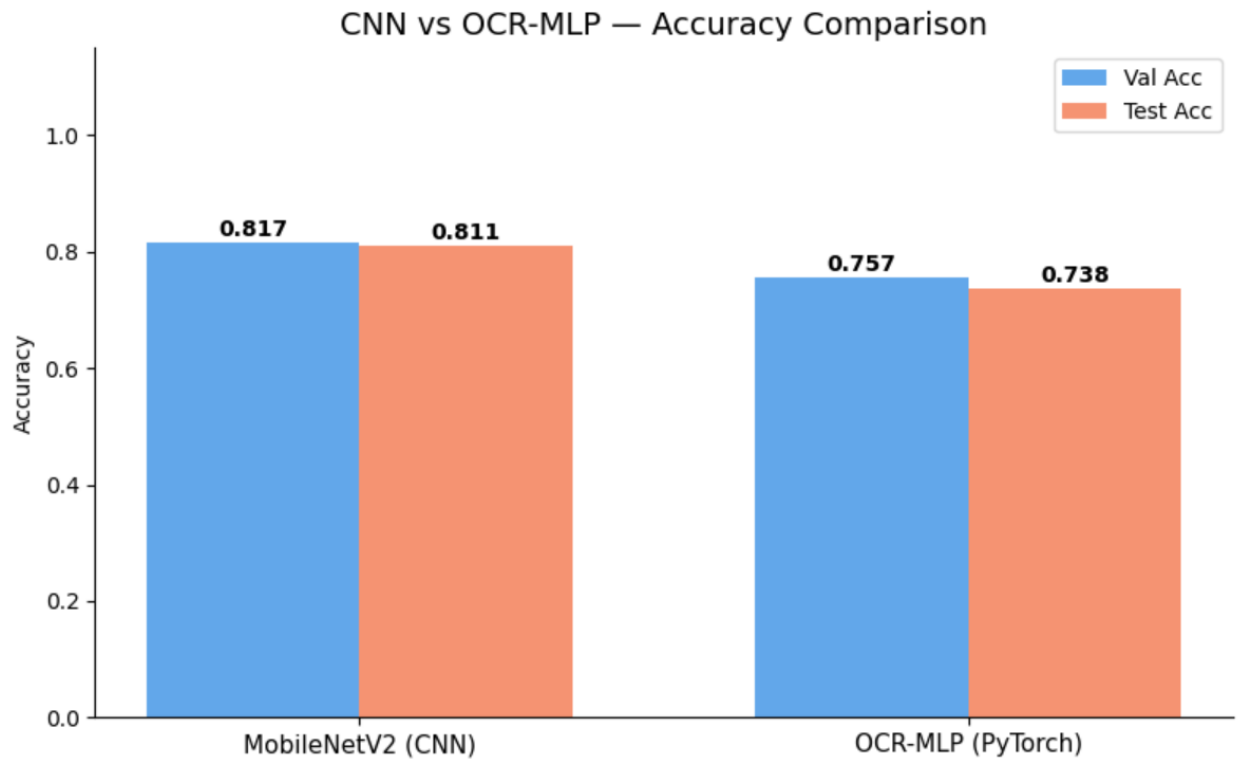


Figure 9: CNN (MobileNetV2) vs OCR-MLP Accuracy Comparison on Validation and Test Sets